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Reactions at graphene|tetracyanoborate ionic liquid interface – New safety mechanisms for supercapacitors and batteries

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ABSTRACT

If an electrochemical energy storage device, which contains tetracyanoborate anions in the electrolyte, is accidentally charged with much higher voltage, it won't explode, because of a self-blocking mechanism. Firstly, voltage (E) more positive than 2.2 V (vs. Ag|AgCl reference electrode) kills the conductivity of single layer graphene and electrochemical reactions stop. Secondly, at $E > 2.2$ V tetracyanoborate anion polymerizes to a dielectric polymer polycyanoborani, which electrically isolates graphite or porous carbon grains from current collectors.

1. Introduction

The advantage of preparing dielectric polymers by electrochemical deposition method can be used in various electronic devices, for example capacitors [1] or transistors [2]. In our previous paper [1] large surface area graphene-based electrodes in 1-butyl-1-methylpyrrolidinium dicyanamide (BMPDCA) ionic liquid were electrochemically coated with a thin polydicyanamide (polyDCA) dielectric layer and used as supercapacitor electrodes with cell voltages up to 10 V.

Safety concerns are mainly connected with Li-ion batteries, which have been reported to sometimes ignite or explode due to inner short-circuit or overcharge [3]. Also a supercapacitor, if accidentally overcharged, would heat up and the formed flammable gasses may cause an explosion. Safer ionic liquids, ionic liquid-ionic liquid mixtures [4], or ionic liquid-solvent mixtures [5] could be used as electrolytes. For example 1-ethyl-3-methylimidazolium tetracyanoborate (EMImTCB) ionic liquid has negligible vapor pressure and thus it is difficult to ignite, it tolerates shortly temperatures up to 400 °C [6,7] and can even function in semisolid forms [8,9]. EMImTCB has been tested as an electrolyte for the supercapacitor by Kurig et al. with cell voltage up to 3.2 V [10–12] and higher capacitance was measured than for similar supercapacitor cells with $EMImBF_4$ ionic liquid electrolyte [10–11]. LiTCB has been tested as an electrolyte for Li-ion batteries and showed better cycling stability than $LiBF_4$ salt [13].

However, there are two new mechanisms connected to electrochemical behavior of tetracyanoborate anion that were not known before. It was briefly described in [14] that nitrile-based ionic liquids polymerize at high positive voltages. Herein, tetracyanoborate-based (TCB) ionic liquid electropolymerization mechanism was studied first time and it is used to dope graphene and some new ideas were generated for increasing the safety of supercapacitors and batteries.

2. Materials and methods

Electrochemical experiments were conducted with 6 mm diameter single-layer graphene or 20 nm thick amorphous carbon (aC) films as electrodes. The process of gluing graphene on a Cu foil onto a glass slide and dissolving the Cu foil has been described in ref. [1]. Amorphous carbon films were deposited onto a glass slide by magnetron sputtering method [15]. Electrochemical measurements were carried out in an argon-filled glove box and ionic liquid EMImTCB (Merck 99.9%) was additionally dried in ultra high vacuum at 110 °C for 24 h. Chlorinated silver wire in EMImTCB was used as the reference electrode (−0.26 V vs. ferrocene/ferrocenium couple [16] or 0.06 V vs. carbon fiber). Electrochemical setup, infrared (IR), Raman, atomic force microscopy (AFM), scanning tunneling microscopy (STM) measurements, and quantum chemical calculations have been described in [15]. Mass spectroscopy data were obtained using gallium ion beam secondary ion mass spectrometer (SIMS) Ulvac-Phi Phi Trift V nanoTOF.

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3. Results and discussion

Single layer graphene was characterized by STM and Raman spectroscopy methods (Fig. 1).

Graphene|EMImTCB interface is nearly ideally polarizable within $-1.85 < \text{voltage } (E) < 1.75$ V. As suggested by phase angles reaching $-87°$, impedance spectra was modelled with simple scheme S1 (inset S1 in Fig. 2a), where nearly constant electrolyte resistance (R_{el} = 11.5 Ω cm^{-2}) was obtained from measurements at graphite and metal|EMImTCB interfaces. Modelled capacitance C_{dl}, voltage (CE) curve (Fig. 2a, black line with round marks) has a characteristic V shape with minimum at 0 V [17].

Graphene resistance R_{gr} vs. E graph (Fig. 2b, black line with round marks) has maximum at 0 V, where the resistance is 8 times higher than at -1.9 V. This behavior has been demonstrated before as graphene field effect transistor with ionic liquid gate has a typical on/off ratio (highest resistance divided by lowest resistance) of 6 [18].

Cyclic voltammetry data in (Fig. 2c inset, black line) show that current starts sharply decreasing at $E < -1.85$ V, which corresponds to reduction of EMIm$^+$ cation to dimer [15,19,20]. There is a small peak at 0.2 V (Fig. 2c inset) – probably corresponding to oxidation of this dimer [15,19,20]. Increase of positive current at $E > 1.75$ V is due to the formation of a polymer, which composition will be discussed later.

Graphene in EMImTCB is inert between $-2 < E < 2$ V; however, when the potential is cycled from 2.2 to -2 V and back (Fig. 2a, blue and green lines with triangular marks), a noticeably different behavior can be observed. Capacitance values are very low between 2.2 V and 0 V and this was modelled with scheme S2 in Fig. 2a, where Warburg diffusion impedance Z_w has been parallelly connected to C_{dl}. At more positive potentials C_{dl} starts rising again reaching similar values as pristine graphene has at these potentials. However, capacitance decreases at $E < -1.6$ V, demonstrating that there is still something at the surface or/and that the graphene is heavily doped. When the voltage starts increasing again the capacitance is very low within -2 and 0 V, thereafter increases to normal values, but decreases again at $E > 1.8$ V. Cyclic voltammetry data (Fig. 2c inset, green dotted line) indicate that no intensive faradaic reactions take place within -2 and 2.2 V, only barely noticeable wide peaks can be detected at -0.2 V and at -1.8 V, which can be explained as absorption and reaction of EMIm$^+$ cation with thin polymeric layer formed at 2.2 V, respectively.

Fig. 2b indicates that the low capacitance values are caused by high resistance values. Interestingly the resistance at 0 V is 273 times higher

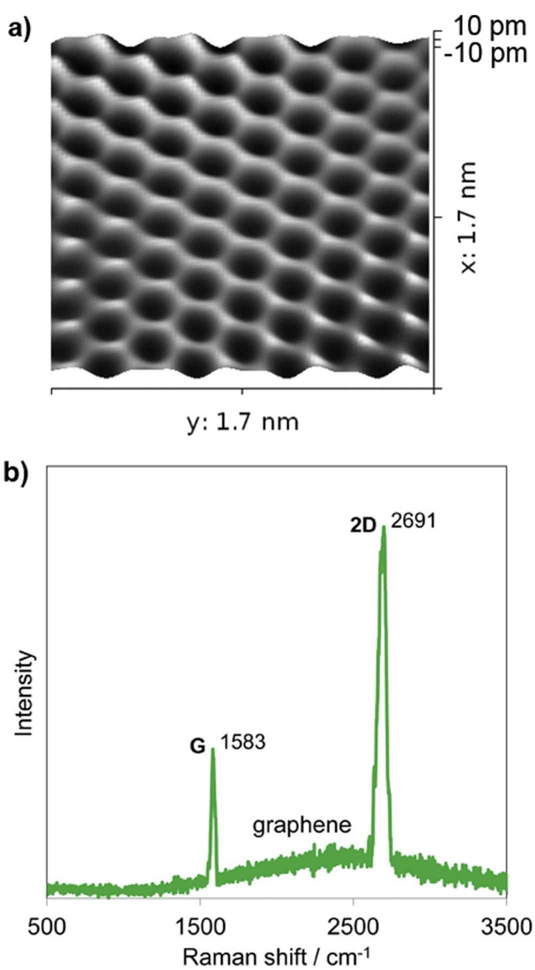

Fig. 1. Characterization of the graphene electrode: a) STM image, b) Raman spectrum (514 nm laser).

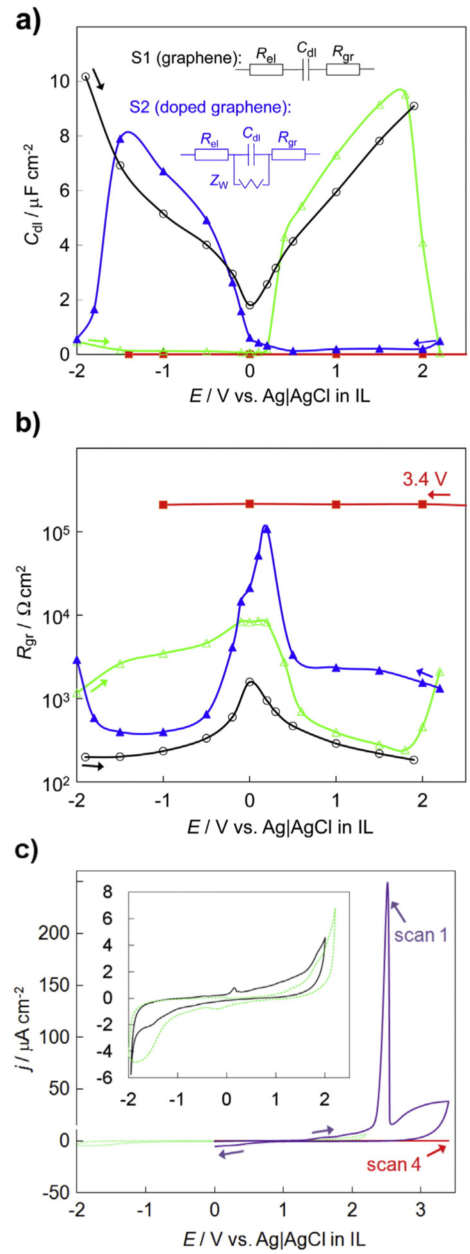

Fig. 2. Impedance data: a) modelled double layer capacitance C_{dl} and b) modelled graphene resistance R_{gr} values at scan rate of 1 mV s^{-1}. Inset in a): used equivalent schemes S1 and S2. c) Cyclic voltammetry data at scan rate of 10 mV s^{-1} for graphene in EMImTCB (inset: smaller voltage range). (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

than at more extreme potentials (or 35 times higher for the −2 to 2.2 V scan direction). This reversible process could be useful in developing graphene-based processors, where established 5–7 times resistance change of pristine graphene-based transistors at room temperature is not enough [18].

When electrode potential increases more positive than 2.2 V, a high current peak develops at 2.4 V (Fig. 2c, scan 1). After holding electrode voltage at 3.4 V, very low current density values can be observed (Fig. 2c, scan 4). Also, capacitance values decrease irreversibly to very low (0.15 nF cm^{-2}) and resistance becomes 146 times higher than that for pristine graphene at 0 V. Probably graphene is so heavily doped that it turns into a dielectric. This process is very different of what takes place at graphene|BMPDCA interface, where the graphene resistance decreases 5 times at 10 V and there is still high capacitance meaning that graphene|BMPDCA system can be used as a high voltage capacitor electrode [1].

Electrochemical data were also obtained using aC films [15] as working electrodes, but behavior characteristic to a typical inert electrode in EMImTCB was found. Differently from graphene, a polymeric layer forms at aC, graphite, or inert metal surfaces when voltage is higher than 2.2 V. The thickness of the layer grows proportionally with applied voltage and there are no changes in the conductivity of the surfaces. After 24 h at 10 V the leaking current is 1 μA cm^{-2}, while 10 times lower values were obtained for the polyDCA [1]. However, in EMImTCB the passivation process is faster and 1 μA cm^{-2} at 10 V is reached in 2 h at 22 °C or in 15 min at 75 °C. Similar polyTCB layers were formed when mixtures of EMImTCB with some solvents (for example propylene carbonate) or ionic liquids (for example EMImBF4) were used as electrolytes.

The formed polymeric layers (polyTCB) are nearly transparent and can be detected on the aC electrode by naked eye as well as under an optical microscope due to different reflective properties (Fig. 3a). The polymer layer is not visible on the graphene electrode. PolyTCB is insoluble in water, acetone, or propylene carbonate, and decomposes only at high temperatures (>250 °C in air and >300 °C in vacuum).

Thickness of 85 nm for the polyTCB film was obtained by scratching a hole into the dielectric film on a stronger aC layer (Fig. 3b), the polyTCB material piled up near the hole. The layer thickness grows linearly with voltage applied 8.5 nm/V, whereas 3.8 nm/V was observed for polyDCA [1]. The surface of polyTCB is relatively flat with root mean square (RMS) roughness value of 1.6 nm. Calculated dielectric (formation) strength and dielectric constant values for polyTCB are 119 MV m^{-1} and 2.9, much lower than obtained for polyDCA (250 MV m^{-1} and 6 [1]).

Mass spectroscopy (MS) data analysis indicate that the polycyanoborani contains carbon ($m/z = +12$), nitrogen (+14, +28), and boron (+10, +11). Mass/charge ratios (Ga ion beam SIMS, 10 s preetch) for positively charged Ga ion beam are 10, 11, 12, 13, 14, 15, 28, 29, 41, 42 and for negatively charged Ga ion beam the values are 12, 13, 26. More detailed interpretation of MS data is complicated due to the relatively low intensities of C_xN_y fragments as trace additions, like absorbed IL, water, and dust cause several orders of magnitude more intensive MS peaks.

In order to analyze the formation of gaseous products, a separate experiment was carried out with a thin transparent cell positioned under an optical microscope. It was found that EMImTCB decomposes without formation of any gas bubbles at the positively (up to 80 V) charged electrode surface.

Raman spectra were measured using 5 different excitation wavelengths, but the peaks in the spectra were undetectably weak and hidden behind strong fluorescence.

We have shown earlier that the structure of organic films electrodeposited onto 20 nm thick carbon electrode [1,15] can be best resolved by comparing experimental IR data with DFT calculated results of ~10 possible products [1,15,19,21,22].

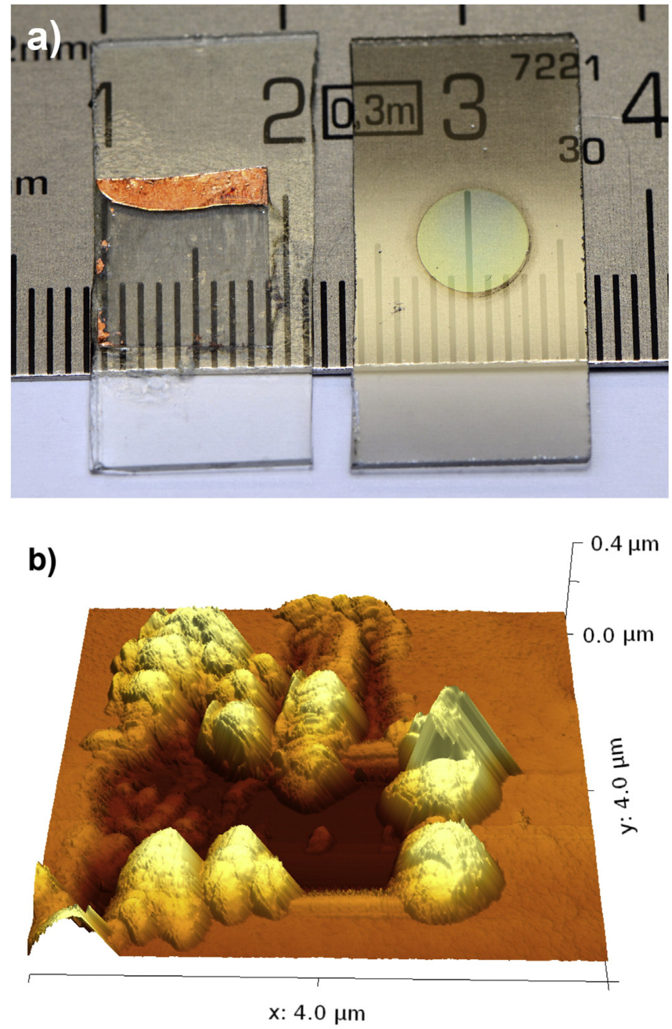

Fig. 3. a) A picture of the graphene (left) and aC (right) electrodes, after being passivated in EMImTCB at 10 V and washed with acetone. b) A scratched AFM image of polyTCB layer on the aC electrode. The 1 × 1 μm rectangular hole has been created by scratching the surface with usual silicon AFM tip using an excessive force in contact mode and thereafter the image was taken with the same tip in non-contact mode.

The infrared spectra "TCB 10 V, 1 V s^{-1}" in Fig. 4a shows absorption bands at 2307, 2228, 2188, 1836, 1552, 1460, 1160, 982, and 944 cm^{-1}, indicating the formation of a new compound at the electrode surface polarized at 10 V. Three vibration bands near 2228 cm^{-1} suggest that there are also 3 different C≡N groups, probably B—C≡N, —C≡N—B, and —C—C≡N. Wide peak at around 1552 cm^{-1} suggests the formation of C—N and N═C bonds. Thus, we propose that polycyanoborani consists $[CN(CN)]_n$ and $[B(CN)_3]_n$ units, where boron has three bonds with carbon and one bond with nitrogen from a neighboring C≡N group (Fig. 4b).

It was found that somewhat different polymer structure can be obtained with lower scan rate synthesis. The spectra "TCB 10 V, 0.01 V s^{-1}" has two clearly separated peaks at 2314 and 2224 cm^{-1} indicating that it is a cyclic structure as the IR spectrum has a good match with calculated cyclic structure "CALC-cyc". Higher synthesis scan rate ("TCB 10 V, 1 V s^{-1}") yielded a mixture of linear and cyclic forms. The cyclic form should be more stable as the calculated energy per $[B(CN)_4]$ unit is 10 kJ mol^{-1} lower than for the linear form and 5-unit cycles have 1 kJ mol^{-1} lower energy than 4-unit cycles.

According to the IR spectra, the formed layer has a different structure than boron carbon nitride prepared by vacuum deposition [23–25], thermal treatment of EMImTCB [7] or chemical synthesis methods [26].

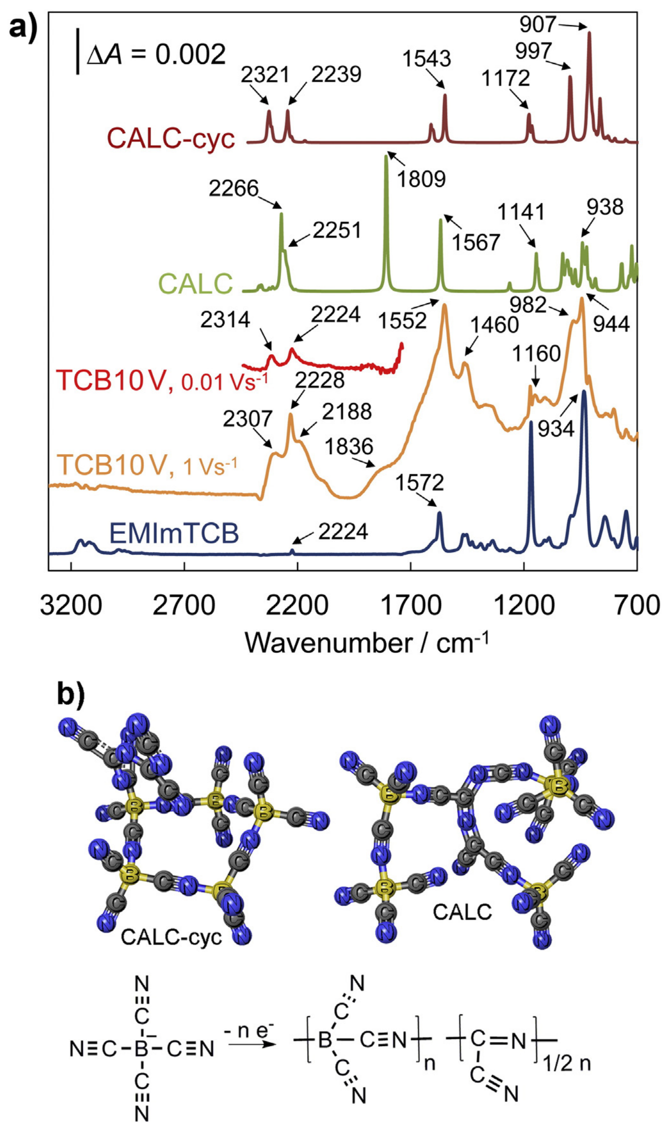

Fig. 4. a) Measured and calculated IR spectra describing the formation of polycyanoborani. b) Proposed reaction mechanism and 3D structures of calculated polymer fragments.

4. Conclusions

The structure of dielectric polymer polycyanoborani, forming in TCB-based ionic liquids at $E > 2.2$ V, has been analyzed, and we demonstrated that a single-layer graphene works in EMImTCB reversibly as a bipolar transistor between -2 and 2.2 V, but becomes irreversibly doped to a dielectric when $E \geq 2.4$ V. Thus, it is proposed that electrolytes containing TCB anions could be used as safe electrolytes because when accidental overvoltage is applied, it kills the conductivity of graphene or forms a dielectric layer and thus electrically disconnects graphite or porous carbon grains from current collectors. Regardless of the given name polycyanoborani, the polymer is actually a nitrile and probably can be handled as safely as nitrile rubber gloves. In addition, EMImTCB ionic liquid has relatively low toxicity to microorganisms [27].

On the other hand, graphene-TCB (or other electropolymerizable anion [14]) system could be used as a simple overvoltage fuse, which disconnects electrical contact when voltage values surpass projected values (~3–5 V).

Acknowledgments

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